

# ECE NUR BAYRAKTAR

## COMPUTER ENGINEERING STUDENT

## CONTACT

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## EDUCATION

### Dokuz Eylül University

B.Sc. Computer Engineering, 3rd Year

### Hasan Türek Anatolian High School

Graduated 2nd in class, 2022

## TECHNICAL SKILLS

**Languages:** Python, Java, C, C#, SQL, JavaScript, TypeScript, HTML, CSS

**AI & Retrieval:** RAG, Phi-3.5 Mini, Foundry Local, Sentence Transformers, BM25, PyMuPDF

**Desktop & Mobile:** PySide6, React Native, Expo, Firebase Realtime Database

**Back End:** Spring Boot, .NET, ASP.NET MVC, REST APIs

**Data & ML:** TensorFlow, Scikit-learn, PostgreSQL, MySQL, MSSQL

**Tools:** Docker, Git, testing, layered architecture

## LANGUAGES

English - Advanced  
German - A2-B1

## INTERNATIONAL EXPERIENCE

### Erasmus+ Exchange, Poland

High-school exchange experience that strengthened cross-cultural communication and adaptability.

## PROFESSIONAL SUMMARY

Third-year Computer Engineering student building privacy-first AI, full-stack, mobile, database, and embedded systems. Developed a local RAG product that combines a polished Windows desktop application with a public product website, translating modern AI research into a practical tool for sensitive document workflows.

## WORK EXPERIENCE

### Software Development Intern

Niso Software Technologies Limited Company | 25 days, 2025

- Supported software development tasks in a professional engineering environment.
- Applied academic programming knowledge to application structure, maintenance, and feature-development workflows.

## PROJECTS

### Local RAG Assistant - Private Document AI

Python, PySide6, Foundry Local, Phi-3.5 Mini, Sentence Transformers, BM25, PyMuPDF

- Built an end-to-end Windows desktop RAG application and product website for private, multi-document question answering with page-level citations; documents and prompts remain local after model setup.
- Combined semantic embeddings with BM25 keyword scoring to retrieve meaning, exact names, numbers, and banking terminology before generating grounded English or Turkish answers.
- Initially positioned for banking teams working with policies, procedures, regulations, and internal knowledge. Designed the architecture to evolve into role-specific assistants for operations, compliance, HR, legal, healthcare, education, and other document-heavy domains.
- Added retrieval evaluation and regression tests, turning the project into a measurable AI product rather than a one-off prototype.

### BloomBook - Book Recommendation Platform

Spring Boot, PostgreSQL, Python, Docker, REST API

- Developed and containerized a full-stack recommendation platform with PostgreSQL persistence and a separate Python recommendation service.

### Dort Bagla Mobile - Realtime Connect Four

React Native, Expo, TypeScript, Firebase Realtime Database

- Built local, online-room, messaging, and offline AI modes with synchronized gameplay and Android distribution builds.

### STM32-Based Command Recognition

Embedded Systems, Machine Learning, STM32, X-CUBE-AI

- Combined signal processing, model experimentation, and embedded deployment constraints for on-device command recognition.

### Additional Software Projects

Spring Boot, .NET, Java, C, XML, XSD, Databases

- Created layered web applications, database-backed CRUD systems, console games, data structures, and schema-validated XML workflows.